

FILE

# STATE OF COLORADO

## DIVISION OF MINERALS AND GEOLOGY

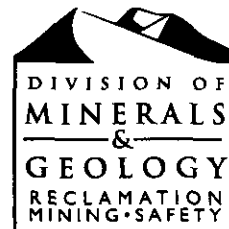
Department of Natural Resources

1313 Sherman St., Room 215

Denver, Colorado 80203

Phone: (303) 866-3567

FAX: (303) 832-8106



February 26, 2003

Mr. Richard Cargill  
St. Vrain Valley Community Watchdogs  
P.O. Box 82  
Hygiene, CO 80533

Bill Owens  
Governor

Greg E. Walcher  
Executive Director

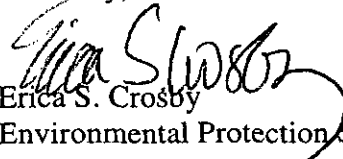
Ronald W. Cattany  
Division Director

RE: CEMEX, Inc., Lyons Mine, Permit No. M-1977-208  
9/5/2002 Inspection Report

Dear Mr. Cargill,

Per your request, please find the 9/5/2002 inspection report for the CEMEX, Lyons Mine, Permit No. M-1977-208. If you have any questions, please do not hesitate to contact me at 303-866-3567.

Sincerely,

  
Erica S. Crosby  
Environmental Protection Specialist

Enclosure



COLORADO DIVISION OF MINERALS AND GEOLOGY  
MINERALS PROGRAM INSPECTION REPORT

PHONE: (303) 866-3567

The Division of Minerals and Geology has conducted an inspection of the mining operation noted below. This report documents observations concerning compliance of the mining operation with the permit and the regulations of the Mined Land Reclamation Board. The report notes 1) Areas of successful compliance; 2) Problems and suggested corrective actions and/or 3) Possible violations to be considered for possible enforcement action by the Mined Land Reclamation Board. **OPERATORS SHOULD READ THIS REPORT CAREFULLY BECAUSE IT MAY REQUIRE CORRECTIVE ACTION AND/OR RESPONSES TO THE DIVISION IN ORDER TO AVOID CONSIDERATION OF POSSIBLE ENFORCEMENT ACTION BY THE MINED LAND RECLAMATION BOARD.**

MINE NAME: Lyons Quarry OPERATOR: CEMEX, Inc.  
COUNTY: Boulder MINERAL: SDG TYPE OF OPERATION: 112  
INSPECTOR(S): Erica Crosby  
MINE ID # OR PROSPECTING ID #: M-1977-208  
INSPECTION DATE: 9/05/02 DATE OF COMPLAINT: 8-12-02  
INSPECTOR'S INITIALS: ESC TIME OF DAY (MILITARY): 10:00  
INSPECTION TYPE CODE<sup>(1)</sup>: MI POST INSP. CONTACTS<sup>(2)</sup>: NO  
JOINT INSP. AGENCY CODE<sup>(2)</sup>: NO REASON FOR INSP. CODE<sup>(3)</sup>: CT  
WEATHER CODE<sup>(4)</sup>: CR BOND CALCULATION TYPE<sup>(5)</sup>: NN  
OP. REP. PRESENT: George Martin; CEMEX, Inc.

1. **INSPECTION TYPE CODE** - [CL-IN: IL=Illegal Operation, MI=Monitoring, MP=Mineral Prospect, SI=Surety-related, PR=Pre-operation]  
2. **POST INSPECTION CONTACTS AND JOINT INSPECTION AGENCY CODE** - [CL-AG: NO=None, BL=BLM, CH=Colo. Dept. Health; CL=Land Board, CT=Citizen; CW=Wildlife, FS=Forest Service, HW=Hwy. Dept., LG=Local Government, SE=State Engr.]  
3. **REASON FOR INSPECTION CODE** - [CL-RS: AG=Other Agency Request, CT=Citizen Complaint, IE=Normal I&E Program, HP=High Priority, PY=Priority]  
4. **WEATHER CODE** - [CL-WE: CL=Cloudy, CR=Clear, IN=Inclement - prevented inspection, RN=Raining, SN=Snowing, WD=Windy]  
5. **BOND CALCULATION TYPE**: - [BC=Complete Bond, BP=Partial Bond, NN=None]

**This list identifies the environmental and permit parameters inspected and gives a categorical evaluation of each. IF PB OR PV IS INDICATED, YOU SHOULD READ THE FOLLOWING PAGES CAREFULLY IN ORDER TO ASSURE COMPLIANCE WITH THE TERMS OF YOUR PERMIT AND APPLICABLE RULES AND REGULATIONS. If PV is indicated, you will be notified under separate cover when the Mined Land reclamation Board will consider possible enforcement action.**

**GENERAL INSPECTION TOPICS**

(AR) RECORDS..... Y (FN) FINANCIAL WARRANTY ..... N (RD) ROADS ..... Y  
(HB) HYDROLOGIC BALANCE..... N (BG) BACKFILL & GRADING ..... Y (EX) EXPLOSIVES ..... Y  
(PW) PROCESSING WASTE/TAILING ..... Y (SF) PROCESSING FACILITIES..... Y (TS) TOPSOIL ..... Y  
(MP) GENL MINE PLAN COMPLIANCE. Y (FW) FISH & WILDLIFE..... Y (RV) REVEGETATION..... Y  
(SM) SIGNS AND MARKERS..... Y (SP) STORM WATER MGT PLAN.... Y (SB) COMPLETE INSP ..... Y  
(ES) OVERBURDEN/DEV. WASTE ..... Y (SC) EROSION/SEDIMENTATION Y (RS) RECL PLAN/COMP... Y  
(AT) ACID OR TOXIC MATERIALS ..... NA (OD) OFF-SITE DAMAGE..... NA (ST) STIPULATIONS..... NA

Y = Inspected and Found in Compliance  
N = Not Inspected

PV = Inspected and Possible Violations Noted  
NA = Not Applicable

PB = Inspected and Problems Noted

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### OBSERVATIONS

This was an inspection of the Lyons Quarry conducted by Erica Crosby of the Division of Minerals and Geology. George Martin of CEMEX, Inc. was present during the inspection.

The purpose of the inspection was to investigate a citizen complaint regarding the Lyons Quarry. The Division received the citizen complaint letter from the St. Vrain Valley Community Watchdogs on August 12, 2002. The letter specifically complains about the fugitive cement kiln dust ("CKD") that was noted blowing off the property on May 14, 2002, July 20, 2002 and July 26, 2002. The letter notes several instances where the sprinkler system was not in operation, thus resulting in plume of CKD off the property. Photographs were included with the letter.

The Lyons cement plant generates approximately 20,000-30,000 tons of CKD annually. The CKD is made up of fined grained particulates of crushed and calcined limestone, shale, gypsum and silica. It is removed from the kiln and mixed with water prior to truck transport to the C-Pit.

Technical Revision 1 ("TR-01") was approved by the Division on February 1, 2000 that specifically addressed disposal of the CKD. The CKD is disposed in the C-Pit and is required to be covered with an interim cap of 2 feet of compacted shale and clay. The working disposal area will be limited to 7 surface acres at 20' lifts. Capping work would normally occur on a 2 to 5 year frequency. The disposed dust is frequently sprayed with water from a number of sprinklers surrounding the disposal area.

The Division inspected the CKD disposal site. At the time of the inspection, the sprinkler system was in operation. According to George Martin, the sprinkler system is on a timer that rotates on 8 sprinklers surrounding the C-Pit. A shale cap was also observed on a portion of the CKD, as required by TR-01.

The Division spoke with Gabi Hoefler of Boulder County Health Department regarding the CKD disposal site and the issues surrounding fugitive dust. Ms. Hoefler noted that over the past several years the operator has made significant progress regarding the CKD disposal. According to Ms. Hoefler, the Air Permit issued to the operator makes an exemption for a fugitive dust emission as a result of winds over 30 MPH. Numerous complains are received at the County Health Department, but seem to fall in the 30 MPH exemption clause.

At this time, the operator is following the CKD disposal plan as noted in TR-01 with the layering of shale on the material and frequent spraying with sprinklers. It appears that the citizen complaint letter is based on dust emissions when the sprinkler system was not in operation. The Division of Minerals and Geology does not regulate fugitive dust emissions. Such activity is generally regulated by the County Health Department and the Colorado Department of Public Health and the Environment; Air Pollution Control Division. The Division ensures the proper disposal of CKD in terms of groundwater protection, closure for final reclamation and what needs to be incorporated into the reclamation cost estimate.

The Division reviewed the previous DMG inspection report dated January 20, 1999. In that report it states the possible necessity of bonding for the cement kiln demolition and disposal. Because the CKD is regulated as a waste generated by the mining operation under the DMG Permit, it would require that a bond be posted as part of the facilities which would need to be demolished and decommissioned following the closure of plant operations. The 1999 inspection reports states that at that time, the company believes that this area of Boulder County is grandfathered as non-conforming use and that the complex and kiln are allowed to remain in place after closure as part of its industrial/commercial final end land use. DMG stated that it would like either written confirmation that the cement kiln is allowed under the current zoning and has the approval of Boulder County to remain in place after reclamation is completed, or a Certificate of Designation regarding the disposal of Cement Kiln Dust. The Division further stated that if neither of these measures are taken in the near future, DMG will recalculate the financial warranty for the site based on the cost of decommissioning the kiln, unless such confirmation is received, in which case the financial warranty will be based on the County requirements for leaving

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the kiln in place. The Division was unable to find such documentation in the permit file.

A review of the permit file shows that the Cement Kiln Facility is within the permit boundary of the Lyons Quarry. The reclamation map shows this area to be reclaimed as "irrigated pasture" as a portion of the post mining land use.

The operator will be required to submit the information noted above by **November 1, 2002**. If this information is not received, then the Division will conduct a follow-up inspection to obtain information to recalculate the financial warranty for the site.

I & E Contact Address

NAME Mr. John Lohr  
OPERATOR CEMEX, Inc.  
STREET P.O. Box 529  
CITY/STATE/ZIP Lyons, CO 80540

☐ CE  
☐ BL  
☐ FS  
☐ HW  
☐ HMWMD (CH)  
☐ SE  
☐ WQCD (CH)

Cc: Carl Mount; DMG (via electronic transmission)

Ms. Gabi Hoefler  
Boulder County Health Department  
3450 Broadway  
Boulder, CO 80304

Mr. Richard Cargill  
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